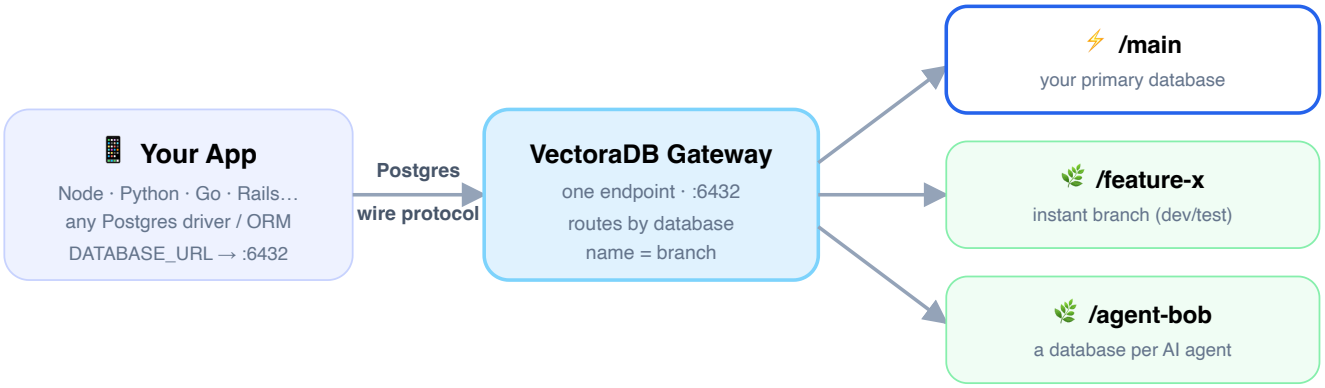
DEVELOPER GUIDE · BUILD AN APP ON VECTORADB

Using VectoraDB in your application

VectoraDB is **PostgreSQL** with three superpowers — instant branches, time-travel, and per-agent databases. Because it speaks the native Postgres wire protocol, **your existing driver, ORM, and SQL work unchanged**. This guide takes you from a clean machine to a running app doing full CRUD.

> **The one idea to hold onto:** VectoraDB gives your app a single, stable Postgres endpoint — `localhost:6432` — and the **database name you connect to is the branch name**. Connect to `/main` for your primary data, or `/feature-x` for an instant, isolated copy. Everything else is ordinary PostgreSQL.

How your application talks to VectoraDB



> **Why this matters for you:** you never manage per-branch ports or hosts. One connection string, and you swap the database name to point your app at production-like data or at a throwaway copy — instantly.

1 Run VectoraDB on your machine

Install the `vdb` command; it sets up everything else (the local Linux VM on macOS, Docker, ZFS, the storage pool, and the image) for you.

macOS

```
# needs Lima for the local VM
brew install lima
curl -fsSL https://raw.githubusercontent.com/\
SauravYadav12/vectoraDB/main/deploy/install.sh | sh
vdb setup
```

Linux

```
curl -fsSL https://raw.githubusercontent.com/\
SauravYadav12/vectoraDB/main/deploy/install.sh | sh
sudo vdb start
```

`vdb setup` (macOS) / `vdb start` (Linux) brings up the database, the single gateway endpoint, and the APIs. Check it:

```
vdb status            # servers + main + branches
vdb stop              # stop everything
```

You now have:

WHAT	WHERE
SQL endpoint (your app connects here)	`localhost:6432`
Web console & dashboard	`make web-dev` → `localhost:5173`
Control API (REST)	`localhost:8080/api`
Agent API (a DB per agent)	`localhost:8088`

2 Create a branch and your schema

Start on `main`, or make an instant branch to work in isolation. Either is a normal PostgreSQL database — create your schema with plain SQL or your migration tool.

Optional: an isolated branch to build in

```
vdb branch create dev          # instant copy-on-write branch of main
vdb branch list                # see branches + their copy-on-write size
```

Create the schema (psql through the gateway)

```
psql "postgres://vectoradb:<API_KEY>@localhost:6432/dev" # or /main

CREATE TABLE notes (
  id          serial PRIMARY KEY,
  title       text NOT NULL,
  body        text,
  created_at  timestamptz DEFAULT now()
);
```

> Prefer a GUI or your ORM's migrations? Point them at the same `postgres://vectoradb:<API_KEY>@localhost:6432/<branch>` URL — VectoraDB is just PostgreSQL, so tools like Prisma Migrate, Alembic, golang-migrate, Flyway, and TablePlus all work.

Connection recap: host `localhost` · port `6432` · user `vectoradb` · password `= your API key` · database `= branch name`. Create a key on the **API keys** page (or `vdb apikey create <email>`).

3 Connect from your application

Put the connection string in an environment variable and use your language's standard Postgres library. Nothing VectoraDB-specific is required in your code.

```
# .env
DATABASE_URL=postgres://vectoradb:<API_KEY>@localhost:6432/dev
```

> **Why `postgres://` ?** Because VectoraDB *is* PostgreSQL — the scheme tells your driver to speak the standard protocol, so every Postgres client and ORM connects with no changes.

> **The password is your API key.** The Gateway (`:6432`) requires a valid key — create one on the **API keys** page (or `vdb apikey create <email>`) and use it as the password.

Node.js — `pg`

```
import { Pool } from 'pg'
const pool = new Pool({ connectionString: process.env.DATABASE_URL })

const { rows } = await pool.query('SELECT now()')
console.log(rows[0])
```

Python — `psycopg` (v3)

```
import os, psycopg
with psycopg.connect(os.environ["DATABASE_URL"]) as conn:
    with conn.cursor() as cur:
        cur.execute("SELECT now()")
        print(cur.fetchone())
```

Go — `pgx`

```
conn, err := pgx.Connect(ctx, os.Getenv("DATABASE_URL"))
if err != nil { log.Fatal(err) }
defer conn.Close(ctx)
```

> **ORMs & frameworks:** Prisma, Drizzle, TypeORM, SQLAlchemy, Django ORM, GORM, ActiveRecord — set their database URL to the same value. To VectoraDB they are ordinary Postgres clients.

4 Create · Read · Update · Delete

The full CRUD cycle — plain SQL on the left, the same operations from Node on the right.

SQL

Node.js (`pg`)

```sql
-- Create
INSERT INTO notes (title, body)
VALUES ('First', 'hello')
RETURNING id;

-- Read
SELECT * FROM notes ORDER BY id;

-- Update
UPDATE notes SET body = 'edited'
WHERE id = 1;

-- Delete
DELETE FROM notes WHERE id = 1;
```

```ts
// Create
const { rows } = await pool.query(
  `INSERT INTO notes(title, body)
   VALUES($1, $2) RETURNING *`,
  ['First', 'hello'])

// Read
await pool.query('SELECT * FROM notes')

// Update
await pool.query(
  'UPDATE notes SET body=$1 WHERE id=$2',
  ['edited', rows[0].id])

// Delete
await pool.query(
  'DELETE FROM notes WHERE id=$1',
  [rows[0].id])
```

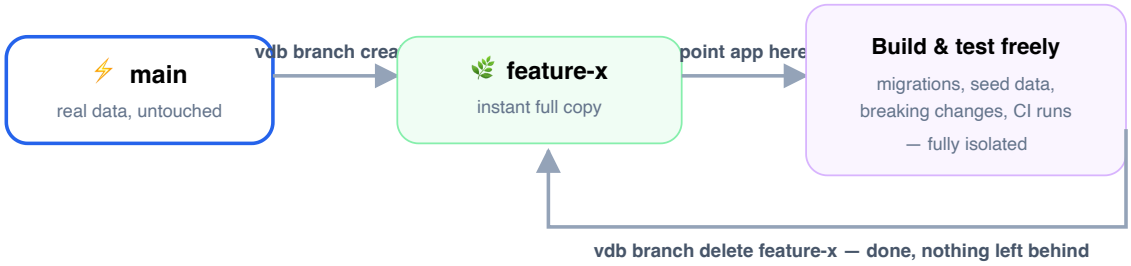
> Use **parameterized queries** (`$1, $2…`) as shown — never string-concatenate user input into SQL. This is standard Postgres safety, and it works identically here.

No console handy? The in-app **SQL Console** (`localhost:5173`) runs these statements against any branch from the browser.

5 The superpower: a branch per feature or test

This is where VectoraDB changes how you build. A branch is an **instant, isolated, full copy** of your database (copy-on-write, so it takes seconds and almost no disk). Point your app at a branch, do anything to it, then throw it away — `main` is never touched.

Branch-per-task workflow




```
vdb branch create feature-x                # seconds, near-zero disk
# in your app / CI:
DATABASE_URL=postgres://vectoradb:<API_KEY>@localhost:6432/feature-x
# …run migrations, tests, a demo, anything…
vdb branch delete feature-x                # throw it away; main is untouched
```

Great for

- **Every pull request** — a preview database with production-like data.
- **CI / integration tests** — a fresh branch per run, deleted after.
- **Risky migrations** — try it on a branch first; if it breaks, delete and retry.
- **Reproducing a bug** — branch, poke at it, discard.

6 One database per AI agent

If your app runs AI agents, give each its own disposable database over HTTP — no SQL admin needed. (These HTTP endpoints require an API key; create one with `vdb apikey create <email>`.)

```
curl -H "Authorization: Bearer $VDB_KEY" \
     -X POST localhost:8088/agents/alice/branch
# -> { "dsn": "postgres://…:PORT/vectoradb" }  — the agent connects to that dsn
curl -H "Authorization: Bearer $VDB_KEY" \
     -X DELETE localhost:8088/agents/alice/branch   # tear it down
```

Quick reference

TASK	COMMAND / VALUE
Start / stop everything	`vdb start` · `vdb stop`
Connection string	`postgres://vectoradb:<API_KEY>@localhost:6432/<branch>`
Create / list / delete a branch	`vdb branch create|list|delete <name>`

TASK	COMMAND / VALUE
Open a SQL shell on a branch	`psql "…6432/<branch>"`
Time-travel (point-in-time restore)	`vdb backup create` · `vdb restore --to latest`
Web console & dashboard	`make web-dev` → `localhost:5173`

That's it. Standard PostgreSQL for your app + instant branches for your workflow. Keep this guide and the live **Docs** / **SQL Console** pages side by side as you build.